

Part A - Source Inventory

File Name	Format	Rows - Columns	Business Owner	Update Frequency	Grain
sales_transactions.csv	csv	10589 - 9	Sales Operations	Daily	One row represents each sale.
customers.csv	csv	186 - 6	CRM	Daily	One row represents the Customer's: ID, Name, Email, Tier, City and the date Joined.
products.xlsx	excel	12 - 5	Product Management	Weekly	One row represents the Product's: ID, its Name, Category, List price and Unit Cost.
branches.csv	csv	6 - 5	Branch Operations	Rarely	One row represents the Branch's: ID, Branch Name, its City, its Province and the date it opened.
employees.xlsx	excel	33 - 7	Human Resources	Monthly	One row represents the Employee's: ID, Name, Branch ID, Role, Hire Date, Daily Rate, Employment Status.
inventory_movements.csv	csv	1541 - 6	Inventory Operations	As Needed	One row represents the movement of the product from its date, its branch, type of movement and quantity.
calendar.csv	csv	537 - 9	Business Planning	Daily	One row represents the date, its year, its quarter, Month number, Month name, year with month, day, day name and day type.

Part B - Business Questions

Manager's Questions	Area	Files Needed	Thought process
Are the Gold tier members visits more often on weekends compared to regular customers?	Sales and Customer Performance	sales_transactions.csv, customers.csv, calendar.csv	I ponder on the frequency of the regular customer visits in the shop on weekends.
Which branches sold the most products per month	Sales Performance	sales_transactions.csv, branches.csv, calendar.csv	I want to know the branch performance.
Which product earns the highest profit margins, also is the item earning the most per unit also the one earning the most overall?	Profitability	sales_transactions.csv, products.xlsx	As a manager i want to know if the best seller is also the one with the highest profit margin.
How many employees does a branch need based on their sales?	Workforce	sales_transactions.csv, branches.csv, employees.xlsx	I noticed that the assigned employees per branch are uneven.
Which of the branches has more wastage of inventory in terms of quantity moved?	Operations	inventory_movements.csv, branches.csv	I want to know which branch waste more of my inventory.
What are the suggestions or feedback of our customers to the products and the shop to better our service?	Customer Feedback	We should have a customer ratings for products and employees service.	Looking at the other small businesses they really get the customers feedback, asking if the items is good and the services etc.

Part C - Data Quality Assessment

File	Column	Dimension	What I observed	Rows affected	What it would break
sales_transactions.csv	Customer ID	Completeness	I notice that there are blank cells in the customer ID	4775	I would not know if they are regular customers or has a tier rank.
customers.csv	Tier	Consistency	I notice that there are two codes for Regulars	16	It would not be consistent in the Tier column as it would split this into two.
customers.csv	JoinedDate	Consistency	The format for the Date is not consistent, stick to YYYY-MM-DD	22	JoinedDate had appear to have two format: YYYY-MM-DD and M/D/YYYY which is not the same if ever it transform to other data.
customers.csv	Email	Completeness	Some Customers has not provided their email	11	It could be use for promotional of the coffee shop.
customers.csv	Customers Name	Consistency	I notice that in the other Customers name has 2 spaces before the name	14	It would break the format as it would not be the same value anymore.
customers.csv	Email	Consistency	I notice that some email are formatted to all-upper case	6	It would not be the same value anymore as there would be an all lower case and all upper case emails.
employees.xlsx	BranchID	Validity	I notice this as there is no B07	1	It would break the referential validity as there is no Branch that has a code for B07
employees.xlsx	EmploymentStatus	Completeness	There are 5 blank cells	5	It can break whether they are full-time or part time
inventory_movements.csv	MovementType	Consistency	I notice that there are 2 format for the delivery	25	It will be two separate value.

Part D - Verdict

At this point in time the data cannot be fully trusted to answer the key business questions as it still need to be accurate and needs prior cleaning. Based on the Data quality assessments it still needs to improve in the completeness, consistency, and validity. If we are going to be using this reports without it being cleaned then we would lead to a wrong business decisions. I would not publish customer tier as there is 4775 missing customer ID in the sales transactions and that would make the tier unreliable. Also the branchID of B07 in the employees data would affect the number of employees and the salary.

If a branch manager demanded and asked for it anyway, I would tell them that using inaccurate date will affect the operationsof the business. It should be cleaned first then i will deliver to them a reliable data that they can actually use to make and improve their business decisions.

AI Log:

who in the business would own it if they use the following file name:

sale_transactions.csv
customers.csv
products.xlsx
branches.csv
employees.xlsx
inventory_movements.csv
calendar.csv

Likely business data owners:

File	Likely owner
sale_transactions.csv	Sales Operations / Finance
customers.csv	Customer Success / CRM / Marketing
products.xlsx	Product Management / Merchandising
branches.csv	Retail or Branch Operations
employees.xlsx	Human Resources / People Operations
inventory_movements.csv	Supply Chain / Inventory Operations
calendar.csv	Finance / Business Planning / Data & Analytics

In a retail setting, a central Data or Analytics team may maintain copies, but these functions would normally own the definitions and accuracy of the source data.

Tool: ChatGPT
Date: August 24, 2026
Purpose: To ask who would own the business if they use the file name mentioned.
Verification: i check in google if this likely owners exist.

how often it would realistically be updated per file?

File	Realistic update frequency
sale_transactions.csv	Real-time or daily; often exported nightly
customers.csv	Daily, with new/changed records throughout the day
products.xlsx	Weekly or as products/prices change; sometimes daily during active promotions
branches.csv	Rarely—only when stores open, close, relocate, or change details; monthly review is typical
employees.xlsx	Daily to weekly, depending on hiring, transfers, and scheduling changes
inventory_movements.csv	Near real-time or daily; warehouse systems commonly produce daily feeds
calendar.csv	Annually for the base calendar, with occasional updates for holidays, fiscal periods, or trading events

For analytics pipelines, a practical default is daily refreshes for sales, customers, employees, and inventory; weekly for products; and monthly or event-driven for branches and calendar data.

Tool: ChatGPT
Date: August 24, 2026
Purpose: To ask how often it would realistically be updated per file.
Verification: its not really accurate as it doesn't have the context of what is in the files. Like for employees, I am the HR so I give it 3 months but to check it monthly.

Using the file name above, help me brainstorm six realistic business questions the way a manager would actually ask them, not report titles. The questions must cover all four functional areas: sales and customer performance, profitability, workforce, and operations. Then for each question, identify which of the seven files would be needed to answer it. At least one question must be one of those files cannot answer. So what data would be needed to answer it.			
Manager's question	Area	Files needed	Gap, if any
"Which branches are falling behind their sales target this month, and is the problem fewer transactions or smaller baskets?"	Sales performance	sale_transactions.csv, branches.csv, calendar.csv	Sales targets are also needed; none of the seven files provides them.
"Which customer groups have stopped buying from us recently, and what did they used to purchase?"	Customer performance	sale_transactions.csv, customers.csv, products.xlsx, calendar.csv	—
"Are the products we're promoting actually making us money after discounts and product costs?"	Profitability	sale_transactions.csv, products.xlsx, calendar.csv	Cannot be answered fully without unit cost/COGS, discount detail, and ideally promotion spend.
"Do we have enough people	Workforce	sale_transactions	A shift schedule,

Tool: ChatGPT

Date: August 24, 2026

Purpose: To help me brainstorm

Verification: I didn't really use these suggestions, I use the format of how he answers it but the coming up of questions is mine as I really think like the business owner and the things I want to know with my business and if it is running successfully.

```
Python
import pandas as pd

sales = pd.read_csv("sale_transactions.csv")

# Show rows where CustomerID is blank, empty, or only spaces
blank_customer_ids = sales[
    sales["CustomerID"].isna() |
    sales["CustomerID"].astype(str).strip().eq("")
]

print(blank_customer_ids)

To see only the number of affected rows:

Python
print(blank_customer_ids.shape[0])

If CustomerID is numeric, the isna() check alone may be enough:

Python
print(sales[sales["CustomerID"].isna()])
```

```
Python
import pandas as pd

df = pd.read_csv("sales_transactions.csv")

blank_customer_ids = df[
    df["CustomerID"].isna() |
    df["CustomerID"].astype(str).strip().eq("")
]

print(blank_customer_ids)
```

```
Python
df["CustomerID"].astype(str).strip().eq("")

print(blank_customer_ids)
```

```
Python
print(blank_customer_ids.shape[0])
```

```
Python
import pandas as pd

customers = pd.read_csv("customers.csv")

# Show records with inconsistent Tier values
inconsistent_tiers = customers[
    customers["Tier"].isin(["Regular", "REG"])
]

print(inconsistent_tiers)

To standardize both formats to one value, for example Regular:

Python
customers["Tier"] = customers["Tier"].replace(
    "REG": "Regular"
)

To check all distinct Tier formats first:

Python
Work with ChatGPT
Ask for approval
```

```
Python
df = pd.read_csv("customers.csv")

# Show records with inconsistent Tier values
inconsistent_tiers = df[
    df["Tier"].isin(["REG"])
]

print(inconsistent_tiers)
```

Python

check to number of REG

```
Python

# Count rows where Tier is exactly REG
reg_count = (customers["Tier"] == "REG").sum()

print(reg_count)
```

For a case-insensitive count, including values such as reg or Reg :

```
Python

reg_count = (
    customers["Tier"]
    .astype(str)
    .str.strip()
    .str.upper()
    .eq("REG")
    .sum()
)

print(reg_count)
```

Work with ChatGPT

Ask for approval

5.6 Terra Medium

Generate Code Markdown Run All Restart Clear All Outputs Python 3.14.2

```
63 C0064 Grace Bautista grace.bautista64@gmail.com REG
67 C0068 Kim Ocampo kin.ocampo68@gmail.com REG
76 C0077 Grace Bautista grace.bautista7@gmail.com REG
85 C0080 Jose Marquez jose.marquez80@gmail.com REG
86 C0087 Dante Navarro dante.navarro87@gmail.com REG
104 C0105 Kim Villanueva kim.villanueva105@gmail.com REG
110 C0126 Miguel Rivera miguel.rivera126@gmail.com REG
127 C0133 Bea Navarro bea.navarro133@gmail.com REG
138 C0139 Jenny Ocampo jenny.ocampo139@gmail.com REG
162 C0163 Cristina Salazar cristina.salazar163@gmail.com REG
169 C0170 Arnel Gonzales arnel.gonzales170@gmail.com REG

City JoinedDate
12 Santa Rosa City 2025-06-28
15 San Pablo City 2025-12-22
23 San Pablo City 2026-01-17
53 Santa Rosa City 2025-03-02
56 Los Banos 2025-11-07
63 Cabuyao City 3/11/2025
...
127 Columbia City 2025-07-25
138 San Pablo City 2025-12-04
162 Santa Rosa City 2025-06-01
169 Binan City 2025-11-30
```

Output is truncated. View as scrollable element or open in a text editor. Adjust call output settings.

```
reg_count = (df["Tier"] == "REG").sum()
print(reg_count)
```

0s Python

16

Python

in customers I want to check the format in the JoinedDate, there are to format: 2026-01-19 and 3/19/2025, count both

```
Python

# Convert to text first, then count each date format
joined_date = customers["JoinedDate"].astype(str).str.strip()

yyyy_mm_dd_count = joined_date.str.match(
    r"^\d{4}-\d{2}-\d{2}$",
    na=False
).sum()

m_d_yyyy_count = joined_date.str.match(
    r"^\d{1,2}/\d{1,2}/\d{4}$",
    na=False
).sum()

print("YYYY-MM-DD format:", yyyy_mm_dd_count)
print("M/D/YYYY format:", m_d_yyyy_count)
```

This counts values such as:

- 2026-01-19 as YYYY-MM-DD
- 3/19/2025 as M/D/YYYY

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Ask for approval

5.6 Terra Medium

Generate Code Markdown Run All Restart Clear All Outputs Python 3.14.2

```
reg_count = (df["Tier"] == "REG").sum()
print(reg_count)
```

0s Python

16

```
# Convert to text first, then count each date format
joined_date = df["JoinedDate"].astype(str).str.strip()

yyyy_mm_dd_count = joined_date.str.match(
    r"^\d{4}-\d{2}-\d{2}$",
    na=False
).sum()

m_d_yyyy_count = joined_date.str.match(
    r"^\d{1,2}/\d{1,2}/\d{4}$",
    na=False
).sum()

print("YYYY-MM-DD format:", yyyy_mm_dd_count)
print("M/D/YYYY format:", m_d_yyyy_count)
```

0s Python

YYYY-MM-DD format: 164
M/D/YYYY format: 22

Python

print("YYYY-MM-DD format:", yyyy_mm_dd_count)
print("M/D/YYYY format:", m_d_yyyy_count)

This counts values such as:

- 2026-01-19 as YYYY-MM-DD
- 3/19/2025 as M/D/YYYY

in the customers check blank email columns

```
Python

# Show customer records where Email is blank, null, empty, or spaces only
blank_emails = customers[
    customers["Email"].isna() |
    customers["Email"].astype(str).str.strip().eq("")
]

print(blank_emails)

# Count them
print("Blank email records:", len(blank_emails))
```

Work with ChatGPT

Ask for approval

5.6 Terra Medium

Generate Code Markdown Run All Restart Clear All Outputs Python 3.14.2

```
# Show customer records where Email is blank, null, empty, or spaces only
blank_emails = df[
    df["Email"].isna() |
    df["Email"].astype(str).str.strip().eq("")
]

print(blank_emails)

# Count them
print("Blank email records:", len(blank_emails))
```

0s Python

CustomerID	CustomerName	Email	Tier	City	JoinedDate
5	C0066	Marites Aquino	NaN	Silver	Santa Rosa City 3/26/2025
9	C0010	Nestor Salazar	NaN	Regular	Los Banos 2025-01-07
20	C0021	Dante Dela Cruz	NaN	Silver	Los Banos 2026-01-09
26	C0027	Alvin Rivera	NaN	Regular	Santa Rosa City 8/23/2025
37	C0038	Ryan Reyes	NaN	Regular	Los Banos 2025-01-23
50	C0051	Rosa Rivera	NaN	Regular	San Pablo City 2025-05-06
68	C0089	Herites Reyes	NaN	Regular	Santa Rosa City 2025-04-23
107	C0108	Ferdinand Santos	NaN	Regular	Los Banos 1/08/2026
120	C0121	Miguel Cruz	NaN	Regular	San Pablo City 2025-11-03
128	C0129	Paolo Ramos	NaN	Regular	Binan City 2025-02-14
158	C0159	Edgar Garcia	NaN	Regular	Columbia City 2025-04-26

Blank email records: 11

Python

in the customer, in column CustomerName there are names that has 2 space before the start of the name ex. " Jenny domingo" i want to count it

```
Python

# Count CustomerName values with exactly two spaces before the name
two_leading_spaces = customers["CustomerName"].astype(str).str.startswith(" ").sum()

print("Names with two leading spaces:", two_leading_spaces)
```

To count any names with one or more leading spaces:

```
Python

leading_spaces = customers["CustomerName"].astype(str).str.match(r"^\s+").sum()

print("Names with leading spaces:", leading_spaces)
```

Work with ChatGPT

Ask for approval

5.6 Terra Medium

Python

Count CustomerName values with exactly two spaces before the name
two_leading_spaces = df["CustomerName"].astype(str).str.startswith(" ").sum()
print("Names with two leading spaces:", two_leading_spaces)

0s Python

Names with two leading spaces: 14

Python

i want to check in the email column the all upper case and the all lower case email

```
Python

# Remove blanks/spaces before checking
email = customers["Email"].dropna().astype(str).str.strip()
email = email[email.ne("")]

# Count emails written entirely in uppercase
upper_case_count = email.eq(email.str.upper()).sum()

# Count emails written entirely in lowercase
lower_case_count = email.eq(email.str.lower()).sum()

print("All uppercase emails:", upper_case_count)
print("All lowercase emails:", lower_case_count)
```

To view those records:

```
Python

upper_case_emails = customers[
    customers["Email"].astype(str).str.strip().str.upper()
    == customers["Email"].astype(str).str.strip()
]
```

Work with ChatGPT

Ask for approval

5.6 Terra Medium

```
check the all upper case in the email column

Python

# Rows where Email is entirely uppercase
all_uppercase_emails = customers[
    customers["Email"].notna()
    & customers["Email"].astype(str).str.strip().ne("")
    & customers["Email"].astype(str).str.strip()
    .eq(customers["Email"].astype(str).str.strip().str.upper())
]

print(all_uppercase_emails[["CustomerName", "Email"]])

# Count
print("All-uppercase emails:", len(all_uppercase_emails))

22:53

Work with ChatGPT
+ Ask for approval 5.6 Terra Medium
```

```
Welcome | Untitled-2.ipynb | Untitled-1.ipynb
Generate + Code + Markdown Run All Restart Clear All Outputs Python
Names with two leading spaces: 14

# Rows where Email is entirely uppercase
all_uppercase_emails = df[
    df["Email"].notna()
    & df["Email"].astype(str).str.strip().ne("")
    & df["Email"].astype(str).str.strip()
    .eq(df["Email"].astype(str).str.strip().str.upper())
]

print(all_uppercase_emails[["CustomerName", "Email"]])

# Count
print("All-uppercase emails:", len(all_uppercase_emails))

CustomerName Email
180 Ryan Reyes RYAN.REYES38@EMAIL.COM
181 Edgar Salazar EDGAR.SALAZAR110@EMAIL.COM
182 Jenny Rivera JENNY.RIVERA130@EMAIL.COM
183 Trisha Garcia TRISHA.GARCIA133@EMAIL.COM
184 Cristina Salazar CRISTINA.SALAZAR118@EMAIL.COM
185 Jenny Dela Cruz JENNY.CRUIZ13@EMAIL.COM
All-uppercase emails: 6

Spaces: 4 LF () Cell 8 of 8
```

```
i want to check in the MovementType column in inventory
movement file the format of "Delivery" and "delivery" and
the count

Python

import pandas as pd

inventory = pd.read_csv("inventory_movements.csv")

# Remove extra spaces before counting
movement_type = inventory["MovementType"].astype(str).str.strip()

delivery_count = movement_type.eq("Delivery").sum()
lower_delivery_count = movement_type.eq("delivery").sum()

print("Delivery:", delivery_count)
print("delivery:", lower_delivery_count)

To display the affected rows:

Python

delivery_rows = inventory[
    movement_type.isin(["Delivery", "delivery"])
]

print(delivery_rows)

count the

+ Ask for approval 5.6 Terra Medium
```

```
Welcome | Untitled-3.ipynb X
Users > ajalcantara > BDAT04_KapeTayoCase > Untitled-3.ipynb > import pandas as pd
Generate + Code + Markdown Run All Clear All Outputs Outline Python
import pandas as pd

df = pd.read_csv("inventory_movements.csv")

# Remove extra spaces before counting
movement_type = df["MovementType"].astype(str).str.strip()

delivery_count = movement_type.eq("Delivery").sum()
lower_delivery_count = movement_type.eq("delivery").sum()

print("Delivery:", delivery_count)
print("delivery:", lower_delivery_count)

Delivery: 1451
delivery: 25

Spaces: 4 () Cell 1 of 3
```

Tool: ChatGPT

Date: August 24, 2026

Purpose: To give me code that i can use in python

Verification: will i trust the python in this ability to count rows affected as long as the code is correct.